

■ Installation Steps — Snort IDS Email & SMS Alerting

Complete guide to install and configure Snort IDS with Python alerting on Ubuntu 22.04

■ Lab Environment

Component	Value
OS	Ubuntu 22.04 LTS
RAM	4 GB
Network Interface	eth0
Snort Machine IP	192.168.1.100
Attacker Machine	Kali Linux — 192.168.1.105

Step 1 — Update System

```
sudo apt update && sudo apt upgrade -y
```

Expected Output:

```
Reading package lists... Done
Building dependency tree... Done
0 upgraded, 0 newly installed
```

Step 2 — Install Dependencies

```
sudo apt install -y build-essential libpcap-dev libpcrc3-dev \
libdumbnet-dev bison flex zlib1g-dev liblzma-dev openssl \
libssl-dev libnghttp2-dev libluajit-5.1-dev pkg-config wget
```

Expected Output:

```
Setting up libpcap-dev (1.10.1-4) ...
Setting up libpcrc3-dev (2:8.39-13) ...
Done.
```

Step 3 — Install DAQ

```
wget https://www.snort.org/downloads/snort/daq-2.0.7.tar.gz
tar -xvzf daq-2.0.7.tar.gz
cd daq-2.0.7
./configure && make && sudo make install
cd ..
```

Expected Output:

```
checking for libpcap... yes
```

Libraries have been installed in: /usr/local/lib

Step 4 — Install Snort

```
wget https://www.snort.org/downloads/snort/snort-2.9.20.tar.gz
tar -xvzf snort-2.9.20.tar.gz
cd snort-2.9.20
./configure --enable-sourcefire
make
sudo make install
cd ..
```

Expected Output:

```
checking for daq_load_modules... yes
make[1]: Leaving directory /home/user/snort-2.9.20
```

Step 5 — Verify Installation ■ [SCREENSHOT]

```
snort --version
```

Expected Output:

```
,,-      -*> Snort! <*-
o" )~    Version 2.9.20 GRE (Build 82)
' ' '    By Martin Roesch & The Snort Team
```

Step 6 — Create Directory Structure

```
sudo mkdir -p /etc/snort/rules
sudo mkdir -p /var/log/snort
sudo mkdir -p /usr/local/lib/snort_dynamicrules
sudo touch /etc/snort/rules/custom.rules
sudo touch /etc/snort/rules/white_list.rules
sudo touch /etc/snort/rules/black_list.rules
```

Expected Output:

```
ls /etc/snort/
rules/

ls /var/log/snort/
(empty for now)
```

Step 7 — Configure snort.conf [SCREENSHOT]

```
sudo nano /etc/snort/snort.conf

# Update these lines:
ipvar HOME_NET 192.168.1.0/24
var RULE_PATH /etc/snort/rules
var WHITE_LIST_PATH /etc/snort/rules
var BLACK_LIST_PATH /etc/snort/rules
```

Step 8 — Add Custom Rules [SCREENSHOT]

```
sudo nano /etc/snort/rules/custom.rules
```

```
alert tcp any any -> $HOME_NET any (msg:"Possible Port Scan"; flags:S; threshold:type threshold, track by_src, count 5,
alert tcp any any -> $HOME_NET 22 (msg:"SSH Brute Force Attempt"; threshold:type threshold, track by_src, count 5,
alert icmp any any -> $HOME_NET any (msg:"ICMP Ping Sweep"; threshold:type threshold, track by_src, count 10, secondaries:
alert tcp any any -> $HOME_NET 80 (msg:"SQL Injection Attempt"; content:"union select"; nocase; sid:1000004; rev:1;
alert tcp any any -> $HOME_NET any (msg:"Nmap SYN Scan Detected"; flags:S; threshold:type threshold, track by_src,
```

Step 9 — Test Configuration ■ [SCREENSHOT]

```
sudo snort -T -c /etc/snort/snort.conf
```

Expected Output:

```
Snort successfully validated the configuration!
Snort exiting
```

Step 10 — Run Snort ■ [SCREENSHOT]

```
sudo snort -A console -q -c /etc/snort/snort.conf -i eth0
```

Expected Output:

```
Running in IDS mode
--== Initialization Complete ==--
```

Step 11 — Install Python Dependencies

```
sudo apt install python3 python3-pip -y
pip3 install twilio
```

Step 12 — Configure Alert Scripts

```
# email_alert.py
EMAIL_SENDER = "your_email@gmail.com"
EMAIL_PASSWORD = "your_app_password"
EMAIL_RECEIVER = "receiver@gmail.com"

# sms_alert.py
ACCOUNT_SID = "your_twilio_sid"
AUTH_TOKEN = "your_twilio_token"
TWILIO_PHONE = "+1XXXXXXXXXX"
YOUR_PHONE = "+1XXXXXXXXXX"
```

Step 13 — Test Full Alert System ■ [SCREENSHOT]

```
# Terminal 1 - Snort running
```

```
sudo snort -A fast -q -c /etc/snort/snort.conf -i eth0
```

```
# Terminal 2 - Alert handler  
python3 Scripts/alert_handler.py
```

```
# Terminal 3 - Simulate attack  
nmap -sS 192.168.1.100
```

Expected Output:

```
[!] Alert detected: Possible Port Scan  
[+] Email alert sent!  
[+] SMS sent! SID: SM123456789
```

■ Installation Complete!

Step	Status
Dependencies installed	■
DAQ installed	■
Snort installed	■
Config validated	■
Custom rules added	■
Alert system working	■

Troubleshooting

Problem	Solution
Snort not found	Run <code>sudo ldconfig</code>
No alerts firing	Check interface with <code>ip a</code>
Email not sending	Enable Gmail App Password
SMS not sending	Verify Twilio credentials
Permission denied	Run scripts with <code>sudo</code>